

HEADACHE AND NEURALGIA

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The IHS (International Headache Society) Classification

A Primary headache disorders

1 Migraine

- 1.1 Migraine without aura
- 1.2 Migraine with aura
- 1.3 Ophthalmoplegic
- 1.4 Retinal migraine
- 1.5 Childhood periodic syndromes that may be precursors to or associated with migraine
- 1.6 Complications of migraine
- 1.7 Migrainous disorder not fulfilling above criteria

2 Tension-type headache

- 2.1 Episodic tension-type headache
- 2.2 Chronic tension-type headache
- 2.3 Headache of the tension-type not fulfilling above criteria

3 Cluster headache and chronic paroxysmal hemicrania

- 3.1 Cluster headache
 - 3.1.1 Cluster headache periodicity undetermined
 - 3.1.2 Episodic cluster headache
 - 3.1.3 Chronic cluster headache
- 3.2 Chronic paroxysmal hemicrania
- 3.3 Cluster headache-like disorder not fulfilling above criteria

4 Miscellaneous headaches unassociated with structural lesion

- 4.1 Idiopathic stabbing headache
- 4.2 External compression headache
- 4.3 Cold stimulus headache
- 4.4 Benign cough headache
- 4.5 Benign exertional headache
- 4.6 Headache associated with sexual activity

The IHS (International Headache Society) Classification

B Secondary headache disorders

- 5** Headache associated with head trauma
 - 5.1 Acute post-traumatic headache
 - 5.2 Chronic post-traumatic headache
- 6** Headache associated with vascular disorders
 - 6.1 Acute ischaemic cerebrovascular disease
 - 6.2 Intracranial haematoma
 - 6.3 Subarachnoid haemorrhage
 - 6.4 Unruptured vascular malformation
 - 6.5 Arteritis
 - 6.6 Carotid or vertebral artery pain
 - 6.7 Venous thrombosis
 - 6.8 Arterial hypertension
 - 6.9 Headache associated with other vascular disorder
- 7** Headache associated with non-vascular intracranial disorder
 - 7.1 High cerebrospinal fluid pressure
 - 7.2 Low cerebrospinal fluid pressure
 - 7.3 Intracranial infection
 - 7.4 Intracranial sarcoidosis and other non-infectious inflammatory diseases
 - 7.5 Headache related to intrathecal injections
 - 7.6 Intracranial neoplasm
 - 7.7 Headache associated with other intracranial disorder
- 8** Headache associated with substances or their withdrawal
 - 8.1 Headache induced by acute substance use or exposure
 - 8.2 Headache induced by chronic substance use or exposure
 - 8.3 Headache from substance withdrawal (acute use)
 - 8.4 Headache from substance withdrawal (chronic use)
 - 8.5 Headache associated with substances but with uncertain mechanism
- 9** Headache associated with non-cephalic infection
 - 9.1 Viral infection
 - 9.2 Bacterial infection
 - 9.3 Headache related to other infection

The IHS (International Headache Society) Classification

- 10** Headache associated with metabolic disorder
 - 10.1 Hypoxia
 - 10.2 Hypercapnia
 - 10.3 Mixed hypoxia and hypercapnia
 - 10.4 Hypoglycaemia
 - 10.5 Dialysis
 - 10.6 Headache related to other metabolic abnormality
- 11** Headache or facial pain associated with disorder of cranium, neck, eyes, ears, nose, sinuses, teeth, mouth or other facial or cranial structures
 - 11.1 Cranial bone
 - 11.2 Neck
 - 11.3 Eyes
 - 11.4 Ears
 - 11.5 Nose and sinuses
 - 11.6 Teeth, jaws and related structures
 - 11.7 Temporomandibular joint disease
- 12** Cranial neuralgias, nerve trunk pain and deafferentation pain
 - 12.1 Persistent (in contrast to tic-like) pain of cranial nerve origin
 - 12.2 Trigeminal neuralgia
 - 12.2.1 Idiopathic trigeminal neuralgia
 - 12.2.2 Symptomatic trigeminal neuralgia
 - 12.3 Glossopharyngeal neuralgia
 - 12.4 Nervus intermedius neuralgia
 - 12.5 Superior laryngeal neuralgia
 - 12.6 Occipital neuralgia
 - 12.7 Central causes of head and facial pain other than tic douloureux
 - 12.8 Facial pain not fulfilling criteria in groups 11 or 12
- 13** Headache not classifiable

History for Headache

- Onset
- Location
- Duration of pain
- Attack frequency & timing
- Pain severity
- Nature of pain
- Aggravating or precipitating factors
- Relieving factors
- Drug Hx
- Social Hx
- Family Hx
- Past headache Hx
- Impact of headache

International Classification of Headache Disorders (ICHD) - II

Primary Headaches

- Migraine
- Tension-type headache (TTH)
- Cluster headache and other trigeminal autonomic cephalgias (TACs). e.g paroxysmal hemicrania
- other primary headaches

Tension-type headache (TTH) – general diagnostic criteria (IHS. The ICHD. 2nd edition (ICHD-II) Cephalgia 2004)

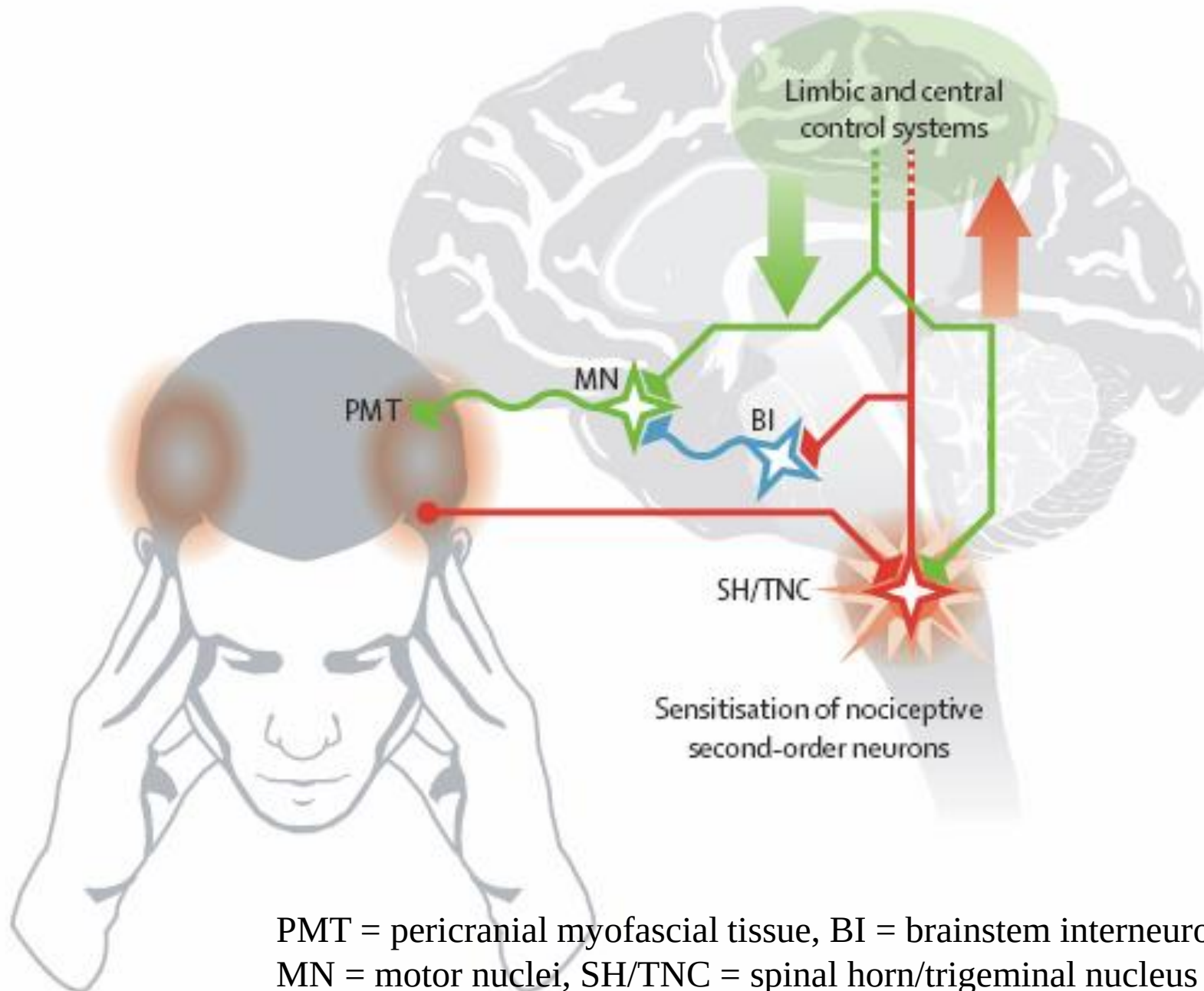
- A. At least 10 attacks for at least 3mo fulfilling B–D
- B. Headache attack lasting from 30min–7d
- C. Headache has at least two of the following characteristics:
 - 1. Bilateral location
 - 2. Pressing/tightening (non-pulsating quality)
 - 3. Mild or moderate intensity
 - 4. Not aggravated by routine physical activity such as walking or climbing stairs
- D. Both of the following:
 - 1. No nausea or vomiting
 - 2. No more than one of photophobia or phonophobia
- E. Not attributed to another disorder

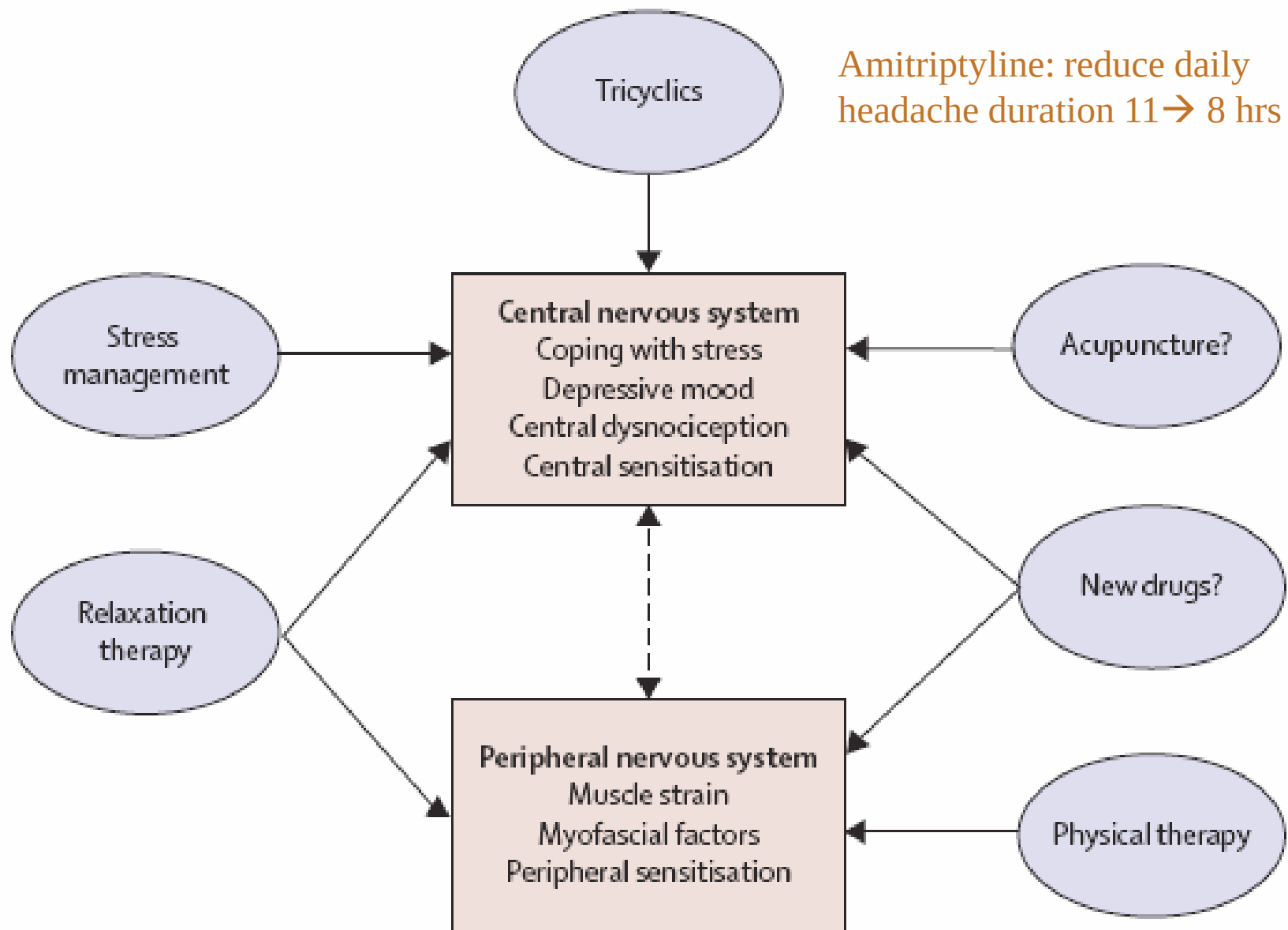
Panel 3: Proposal for stricter diagnostic criteria for TTH

- B Headache lasting from 30 min to 7 days**
- C At least three of the following pain characteristics:**
 - 1 Bilateral location
 - 2 Pressing or tightening (non-pulsating) quality
 - 3 Mild or moderate intensity
 - 4 Not aggravated by routine physical activity such as walking or climbing stairs
- D No nausea, vomiting (anorexia can occur), photophobia, or phonophobia**
- E Not attributed to another disorder**

From the 2nd edition of the International Classification of Headache Disorders (ICHD-II) appendix A2.²

A model for pathophysiology of CTTH





Putative pathophysiological targets of preventive therapies for TTH

TREATMENT OF TTH

Aim: prevent TTH from becoming CHRONIC

Short-term, abortive treatment of attacks (mainly pharmacological)

- ✗ NSAIDs (ibuprofen, naproxen, aspirin 500mg/1000mg)
- ✗ COX-2 inhibitor (lumiracoxib)
- ✗ Paracetamol (500mg/1000mg)
- ✗ combination (NSAID, sedative, caffeine, tranquillisers)
- ✗ Topical Tiger Balm or peppermint oil

LONG-TERM PROPHYLACTIC TREATMENTS FOR TTH

Non-pharmacological (stress management, relaxation therapy, physical therapy)

Pharmacological

Antidepressants

- ✗ amitriptyline, maprotiline, mirtazapine
- ✗ slow release venlafaxine

Alpha2- adrenergic agonist

- ✗ tizanidine (+ amitriptyline)

Anti-epileptic drug

- ✗ topiramate

Migraine

- a neurovascular disorder of the brain
- common, chronic, incapacitating
- **attacks of**

1. severe headache

2. autonomic nervous system dysfunction

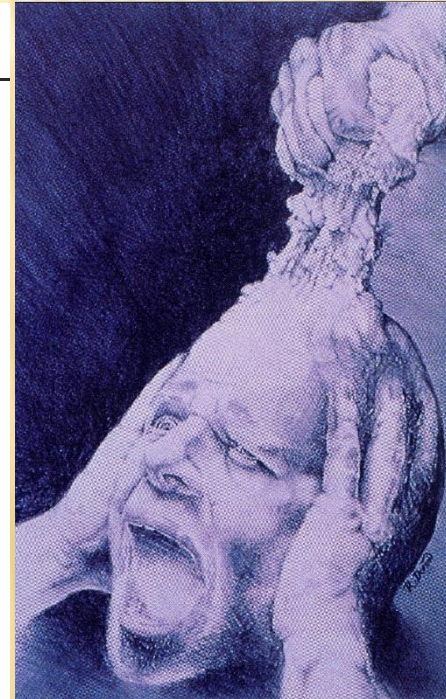
3. in some, aura involving neurologic Sx

International Classification of Headache Disorders – II

ICHD-II criteria for migraine without aura. The form is considered probable when episodes fulfil all except one of criteria A-D

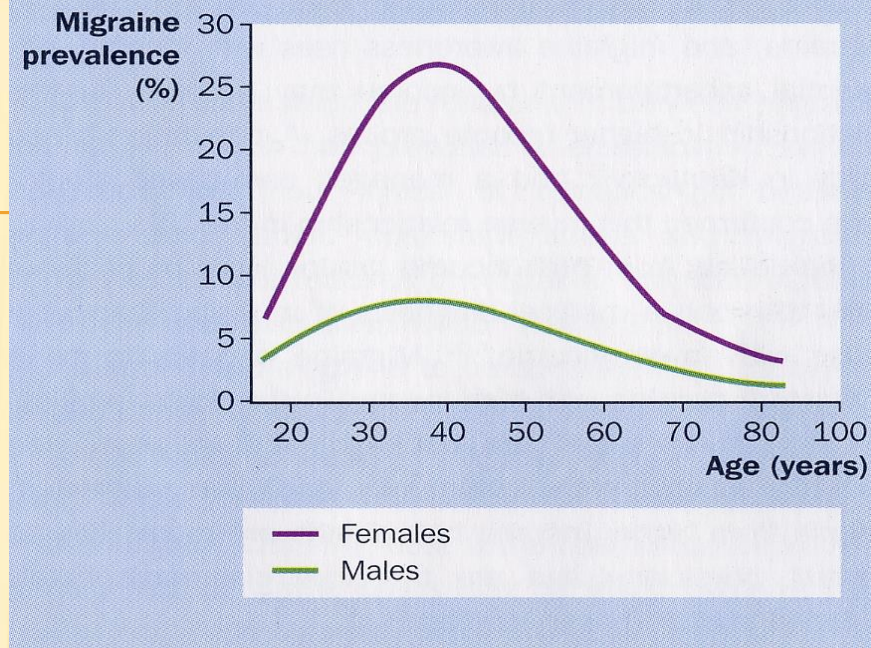
Migraine without aura

- A. At least five attacks fulfilling B–D
 - B. Headache attack lasting 4–72h^a
 - C. Headache has at least two of the following characteristics:
 - 1. Unilateral location
 - 2. Pulsating quality
 - 3. Moderate or severe intensity
 - 4. Aggravation by or causing avoidance of routine physical activity such as walking or climbing stairs
 - D. During headache, at least one of the following:
 - 1. Nausea and/or vomiting
 - 2. Photophobia and phonophobia
 - E. Not attributed to another disorder
-



^aIn children younger than 15y, attacks may last 2–48h. ICHD-II,

MIGRAINE - PREVALENCE



- ✗ 10% general population (6% males, 15% females) active migraineurs; 2010 global prevalence 14.7%
- from age 16 yr, 2–3 times more frequent in female
- Peak incidence around age 14-16 yr for female
- peak incidence at age 10-12 yr for male
- in women a less abrupt second rise until age 40 yr
- females ~age 40 yr: highest prevalence (24%)
- onset before age 50 yrs in 90%

MIGRAINE

Two main types:

1. migraine without aura (75%)
2. migraine with aura (1/3)

Migraine patients defined as having

- ≥ 2 attacks with aura or ≥ 5 attacks without aura

Aura symptoms: 1 or more transient focal neurological aura Sx

1. *visual (99% of patients), together with*
 2. *sensory (31%) Sx or*
 3. *aphasic (18%) Sx and*
 4. *rarely motor Sx (6%)*
- ✗ gradual development of aura Sx over >4 min, or several Sx in succession
 - ✗ usually last 5-60 min (motor Sx longer)
 - ✗ aura Sx nearly always precede headache (follows or accompanies aura within 60 min)

up to 33% migraineurs experience both types of attack

up to 42% may have **migraine aura without headache**

LASHLEY'S AURA



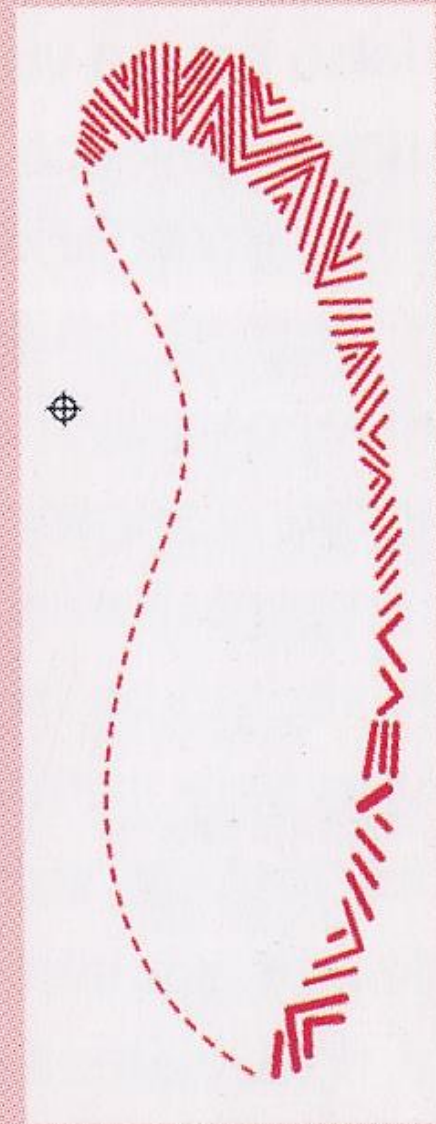
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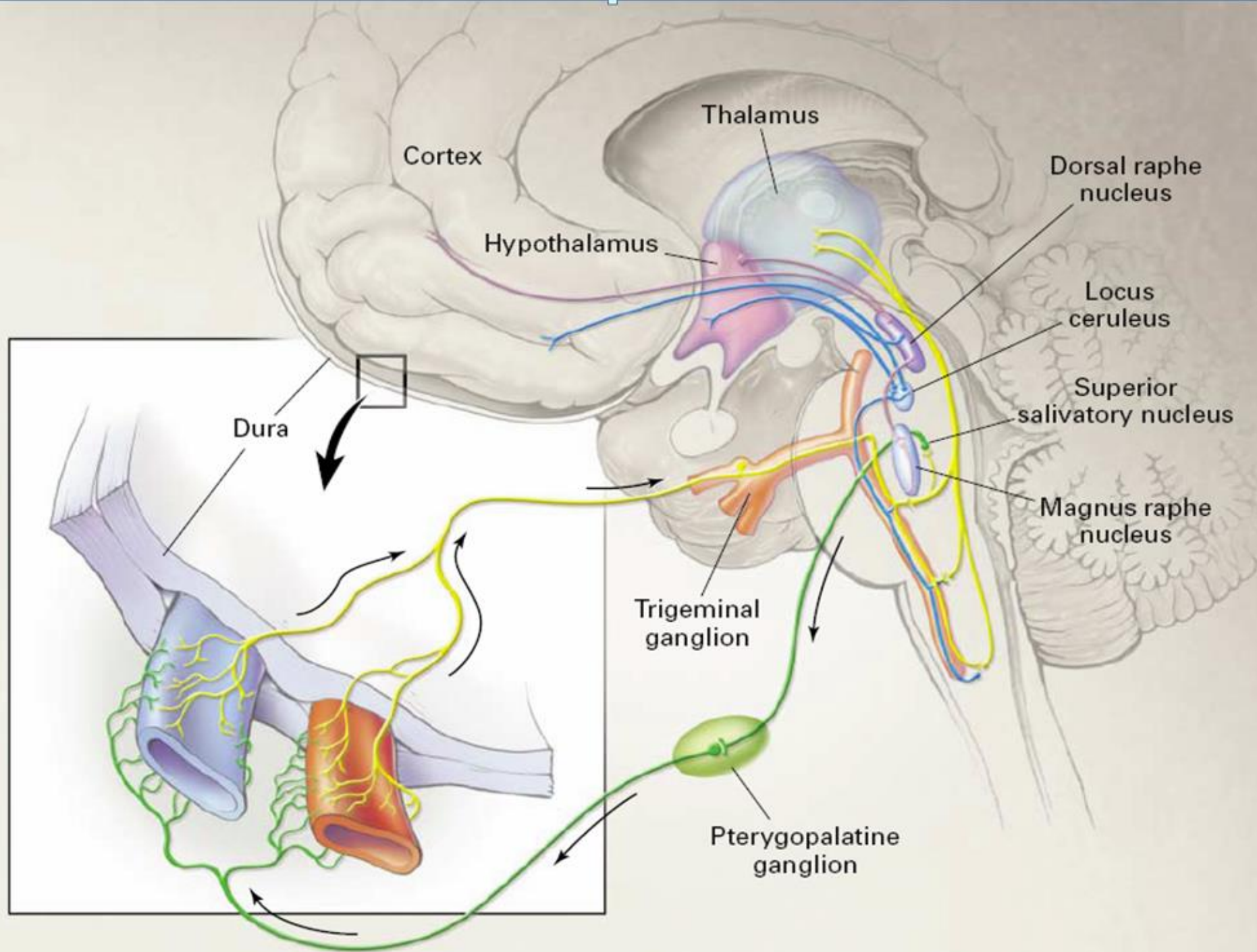


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MIGRAINE - MECHANISMS

Dysfunction of brainstem pathways that normally modulate sensory input:

- ✗ Trigeminovascular input from meningeal vessels
- ✗ Trigemino-autonomic reflex (exaggerated in trigeminal-autonomic cephalgias, TAC)
- ✗ Modulation of trigeminovascular nociceptive input from dorsal raphe nucleus, locus ceruleus, magnus raphe nucleus
- ✗ Reduced cerebral blood flow (oligemia) during migraine aura, which is preceded by a phase of focal hyperemia



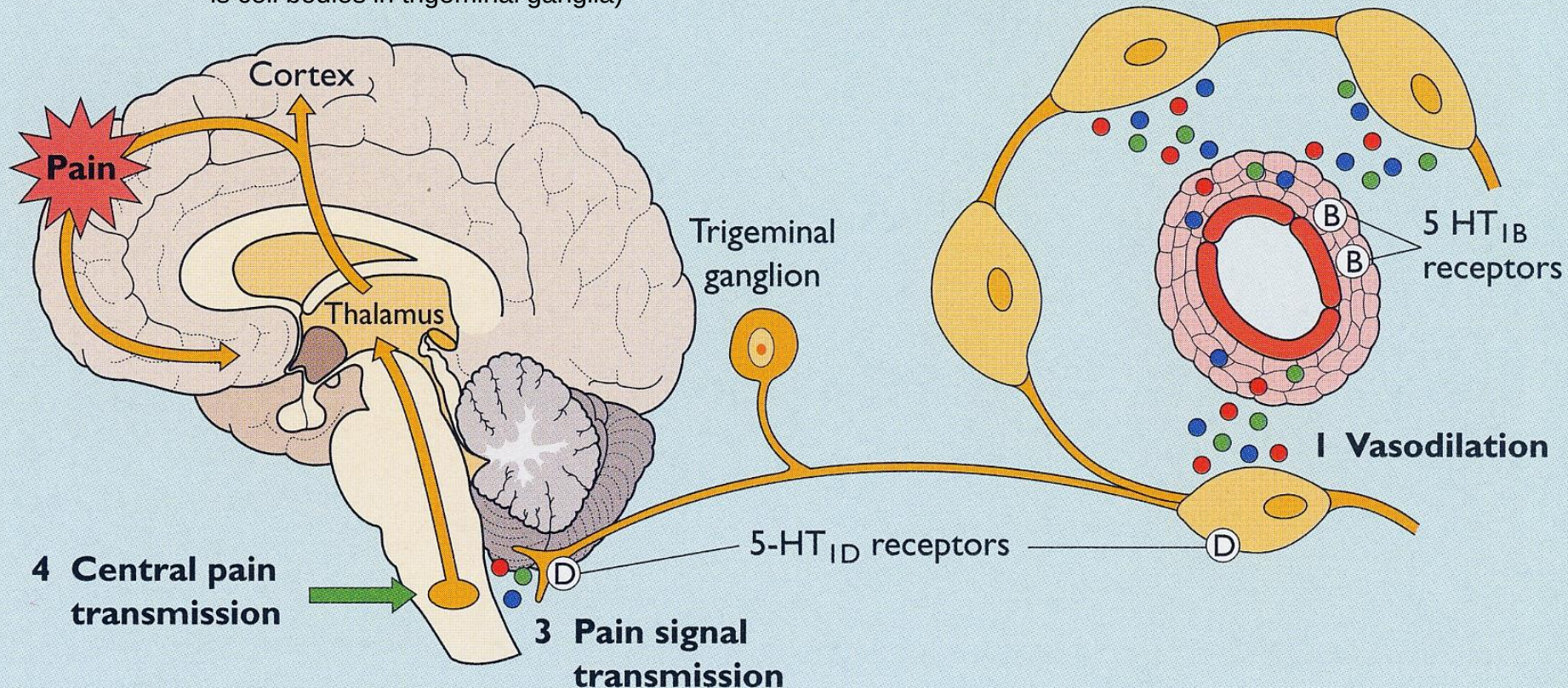
Pathophysiology of migraine

Neuropeptides

- Neurokinin A
- Substance P
- CGRP (main source of calcitonin gene-related peptide is cell bodies in trigeminal ganglia)

2 Neuropeptide release

- vasodilation
- neurogenic inflammation



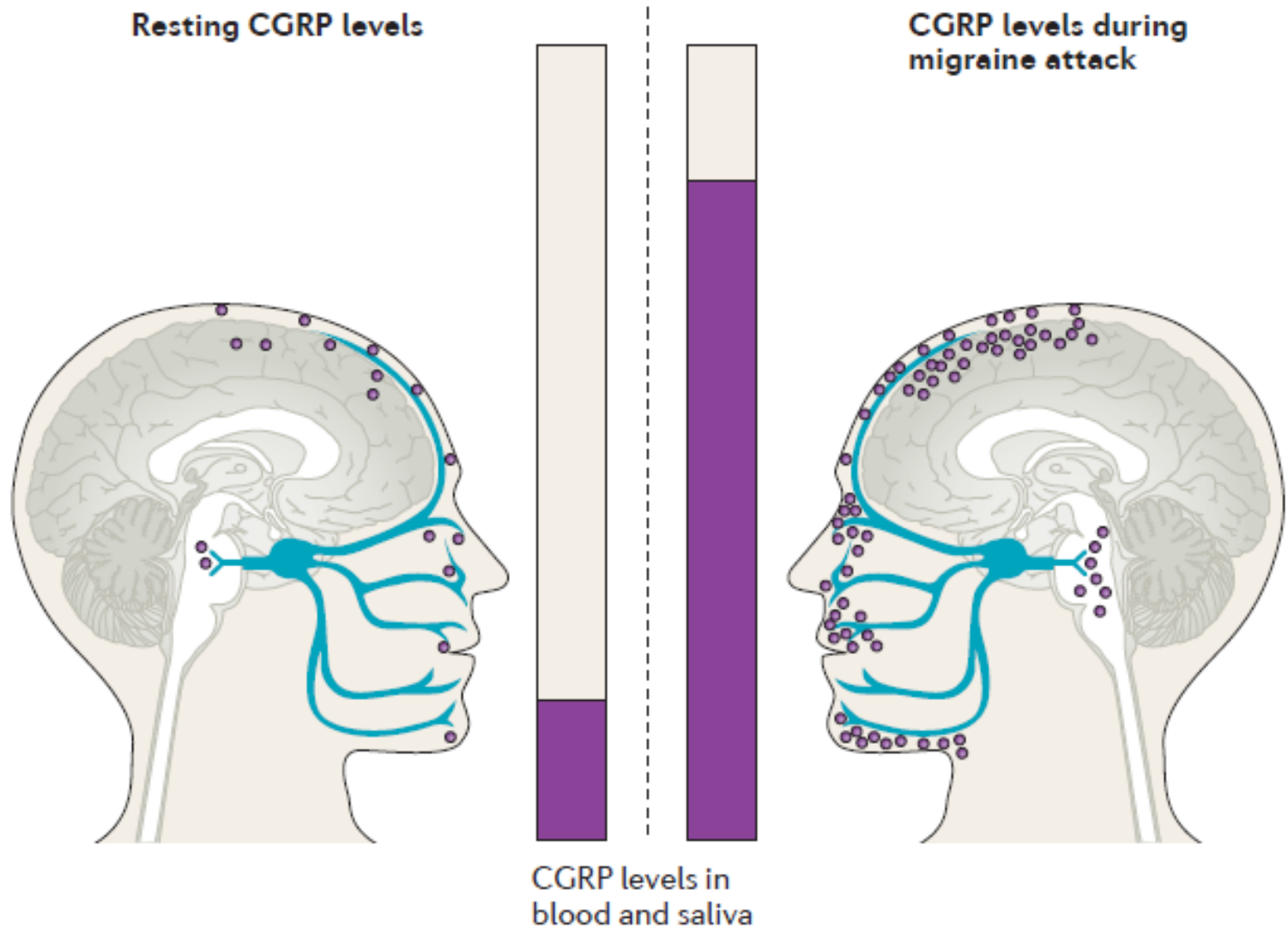
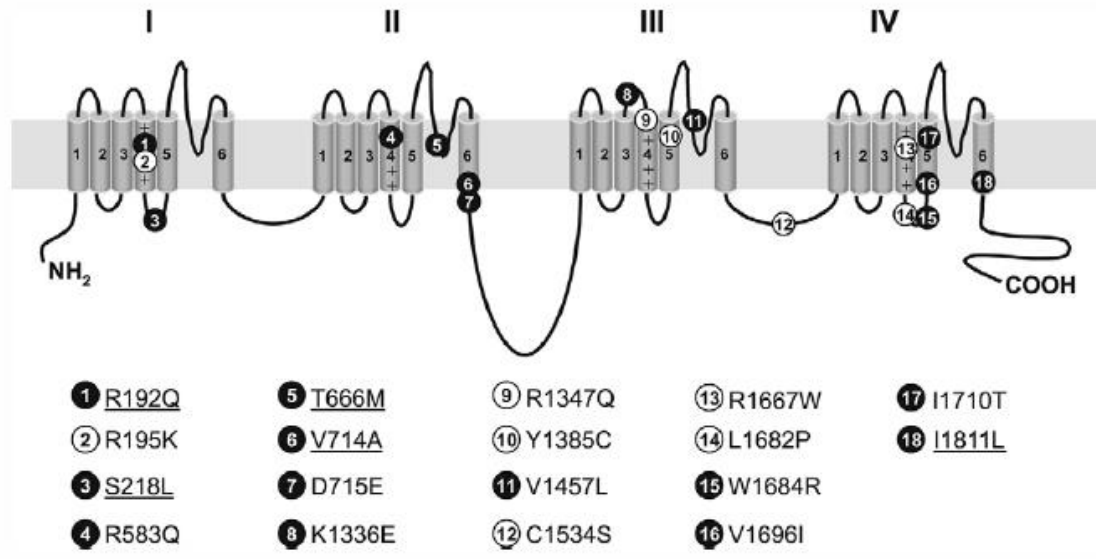
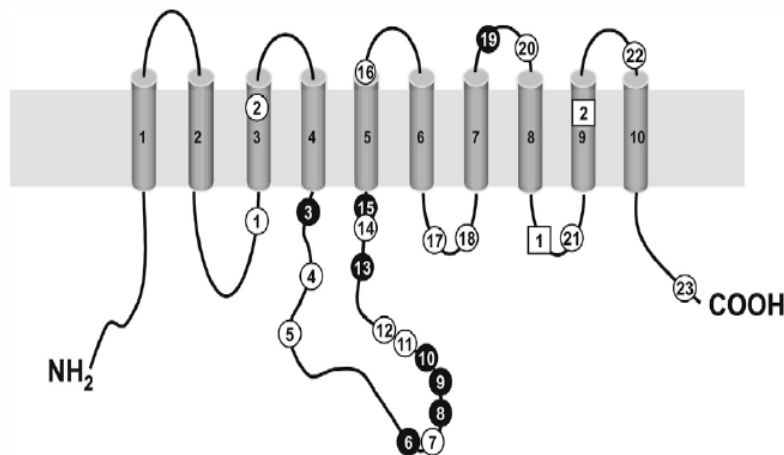


Figure 1 | CGRP levels during migraine attacks. Calcitonin gene-related peptide (CGRP) is released from trigeminal sensory afferents and the spinal trigeminal nucleus. During migraine attacks, CGRP levels in the blood and saliva are increased.

FAMILIAL HEMIPLEGIC MIGRAINE

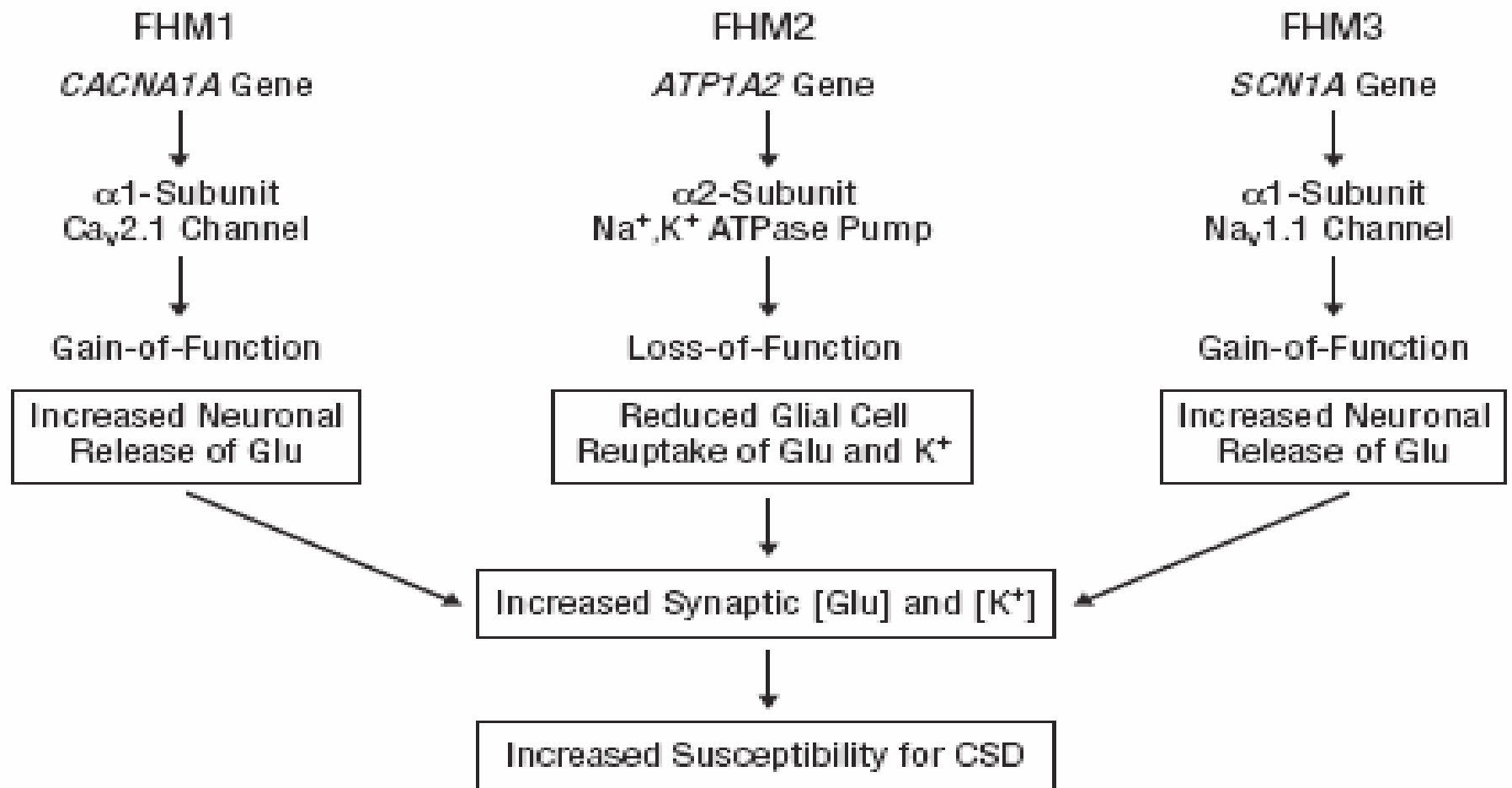


Familial hemiplegic migraine (FHM) 1 caused by mutations in different sites of voltage gated calcium channel $\alpha 1$ subunit



FHM 2 caused by mutations of Na^+, K^+ -ATPase $\alpha 2$ subunit

- | | | | | |
|---------|---------|---------|---------|-------------------|
| ① T263M | ⑥ R593W | ⑪ E700K | ⑰ P796R | ⑳ R937P |
| ② G301R | ⑦ A606T | ⑫ D718N | ⑱ M829R | ㉑ P979L |
| ③ T345A | ⑧ G615R | ⑬ M731T | ⑲ R834Q | ㉒ X1021R |
| ④ T376M | ⑨ V628M | ⑭ R763H | ㉑ W887R | ① del 5aa ins I |
| ⑤ R383H | ⑩ R689Q | ⑮ L764P | ㉒ E902K | ② del CT (S966fs) |



SCN = sodium channel, Glu = glutamate (excitatory), CSD = cortical spreading depression

FHM genes cause increased neuronal excitability and susceptibility to cortical spreading depression

All 3 genes are implicated in ION TRANSPORT

MIGRAINE: TREATMENT

TREATMENT OF MIGRAINE: ACUTE ATTACK

1. Simple analgesics

	Half-life (h)	Dose (mg)	Number needed to treat (2 h pain relief)
Aspirin			
Tablet, intravenous, or combined with caffeine and paracetamol	0.25	975-1000	4.9
Salicylate (active form)	5-6 (after 1 g dose)	975-1000	3.3*
Ibuprofen			
Tablet	2	400	3.2
Naproxen			
Tablet	14	500-550 (up to 825 mg)	7.0
Diclofenac potassium			
Tablet	2	50	6.2
Powder for oral solution	2	50	4.5

2. Ergot derivatives: ergotamines/dihydroergotamines

- 5-HT receptor agonist, inhibit peptide release
- sustained generalized vasoconstrictor effects → adverse vascular events
- high risk of overuse syndromes & rebound headache

3. Ditans (lasmiditan): 5-HT 1F serotonin receptor antagonist

4. Gepants (small molecule CGRP receptor antagonist)

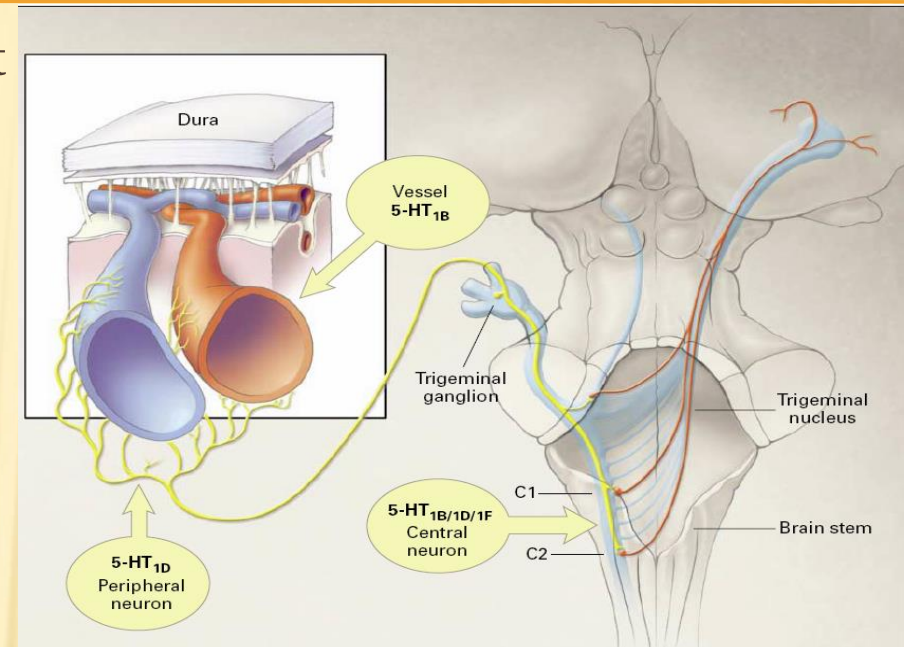
- for migraine attack with moderate or severe headache

5. Anti-emetic drugs/pro-kinetic agents - metoclopramide

TREATMENT OF MIGRAINE: ACUTE ATTACK

6. Triptans

- ✖ selective pharmacology; simple, consistent pharmacokinetics
- ✖ evidence-based prescription instructions
- ✖ moderate side effects (vasoconstriction, contraindicated in vascular disorders e.g. coronary artery disease, cerebrovascular disease, PVD, uncontrolled hypertension)
- ✖ **Serotonin 5-HT_{1B/1D} receptor agonists:**
 - potential mechanisms of action
 1. **cranial vasoconstriction**
 2. **peripheral neuronal inhibition**
 3. **inhibition of transmission through second-order neurons of trigeminocervical complex** → inhibit effects of activated nociceptive trigeminal afferents



TREATMENT OF MIGRAINE: ACUTE ATTACK

- ✘ Limited to 2-3 days per week: avoid analgesic OVERUSE which can lead to medication overuse headache (MOH)
- ✘ Keep headache diary: monitor for escalation in drug use
- ✘ **Avoid opioids – mask pain without suppressing pathophysiologic mechanism of attack; can cause cognitive impairment/ addiction**

Migraine – Prophylactic treatment

Indicated when:

- attacks weekly/several times a month
- ≥ 8 headache days per month
- attacks less often but very prolonged & debilitating
- hemiplegic migraine, migraine with brainstem aura
- frequent, prolonged, uncomfortable aura Sx
- migranous infarction

most effective prophylactic agent available

typically reduce headache frequency by $\geq 50\%$ in
~50% patients

Drugs for Migraine Prophylaxis: proven or well accepted

Drug	Selected Side Effects
Beta-blockers (propranolol, metoprolol)	Reduced energy (lethargy), tiredness, depression, postural Sx; contraindicated in asthma
Antidepressants (amitriptyline & other TCA; venlafaxine)	Drowsiness, anticholinergic side effects, weight gain; serotonin syndrome
Calcium channel blockers (flunarizine, verapamil)	Tiredness, weight gain, depression, Parkinsonism
Valproate	Drowsiness, weight gain, tremor, alopecia, fetal abnormalities, thrombocytopenia, hepatotoxicity
Topiramate	glaucoma, paresthesia, weight loss, increased risk of renal stones, pregnancy & lactation
Gabapentin	Dizziness, sedation
Methysergide (serotonergic agent)	Drowsiness, leg cramps, alopecia, retroperitoneal fibrosis, mediastinal fibrosis, cardiac valve fibrosis
Disulfiram (anticonvulsant agent)	Drowsiness, weight gain

Drugs for Migraine Prophylaxis: recently accepted

Drug	Selected Side Effects
OnabotulinumtoxinA (RCTs : reduced headache frequency, intensity & no. of triptan doses; improved QoL in chronic migraine patients): IM, SC; inhibition of neuropeptides or neurotransmitter release	muscle weakness and atrophy
CGRP or CGRP receptor antagonist (monoclonal antibodies): SC, IV; neutralize circulating neuropeptides or CGRP receptors	Cautions and contraindications: hypersensitivity, heart disease, cerebrovascular disease, inflammatory bowel disease

NON-PHARMACOLOGICAL TREATMENTS

Neurostimulation devices

- ✗ Transcranial magnetic stimulation approved for
 1. acute treatment of migraine with aura in 2014
 2. acute and prophylactic treatment of migraine headache
- ✗ External trigeminal nerve stimulation approved for acute treatment of migraine with/without aura & preventive treatment of episodic migraine

MIGRAINE

Migraine (-like) symptoms may be caused by:

- ✗ *arteriovenous malformations*
- ✗ *internal carotid dissection*
- ✗ *epilepsy*
- ✗ *mitochondrial DNA disorders (mitochondrial encephalomyopathy with lactic acidosis and stroke-like episodes [MELAS])*
- ✗ *cerebral autosomal dominant arteriopathy with subcortical infarction and leucoencephalopathy (CADASIL)*

HEADACHE – SECONDARY: DIAGNOSIS OF PRIMARY HEADACHE NEEDS CAREFUL EXCLUSION OF SECONDARY CAUSES OF HEADACHE

Temporal arteritis (TA), typically elderly: symptoms:

- Headache (72%), typically bitemporal
- Polymyalgia rheumatica (58%)
- Malaise, fatigue (56%)
- Jaw pain during chewing (jaw claudication) (40%)
- Fever (35%)
- Cough (17%)
- Neuropathy (14%)
- Sore throat , dysphagia (11%)
- Amaurosis fugax (10%), TIA/stroke (7%)
- Permanent visual loss (8%) **ophthalmological Cx, mostly optic neuropathy, ~20%**
- Limb claudication (8%)
- Neuro-otological disorder (7%)
- Scintillating scotoma (5%)
- Tongue claudication (4%)
- Depression (3%)
- Double vision, diplopia (2%)
- Myelopathy (0.6%)

Diagnostic criteria for temporal arteritis (TA)

- Age at onset ≥ 50 years
- New headaches
- Abnormalities of temporal artery at clinical examination
- Raised ESR (≥ 50 mm/hr)*
- Abnormal findings on biopsy of temporal artery

N.B. 3 or more criteria: Dx of TA with 91.2% specificity & 93.5% sensitivity.

**Raised C-reactive protein (>5 mg/L) is another, currently used, clinically useful laboratory parameter*

Neuralgia

- sudden, intense, sharp, aching, lancinating, burning, stabbing pain lasting seconds to < 2 min
- recurring repeatedly within short periods
- triggered by sensory/mechanical stimuli

TRIGEMINAL NEURALGIA

- ✖ most frequent facial neuralgia, prevalence : 3–6 per 100 000
- ✖ incidence ↑ with age, ~70% patients >60 yrs at onset
- ✖ clinical hallmark: paroxysms of pain, very intensive, almost exclusively unilateral in a trigeminal distribution
- ✖ typical: in second or third division (~5% first division)
- ✖ ~3%: pain bilateral suggesting a secondary origin
- ✖ spontaneous or precipitated by sensory stimulation of certain areas in face, e.g. cheek, chin, lip, tongue; trigger spots
- ✖ touching/washing face, shaving, brushing teeth, chewing: typical triggers
- ✖ stereotypic pain attacks usually many times per day, for wks or mths, then suddenly stop, subsequent pain-free periods for mth or yr
- ✖ disease course: pain attacks tend to become more frequent over yr, & remission periods shorter

Trigeminal Neuralgia (TN)

Idiopathic TN (primary TN): normal neurological & MRI examination → no cause detectable

Most common identifiable cause - compression of trigeminal nerve root by an aberrant loop of a blood vessel (60–90%) in neurosurgical and neuroradiological series

MRI for detection of neurovascular contact:

- sensitivity (50–95%)
- specificity (65–100%)
- false +ve (7–15%)
- false -ve (10%)

TRIGEMINAL NEURALGIA (TN)

up to 15% patients, an underlying cause can be identified
(**symptomatic TN**):

- ✗ multiple sclerosis especially young patients, bilateral
- ✗ other structural lesions mainly in pontine region:
 1. *vestibular/trigeminal schwannoma, meningiomas*
 2. *epidermoid or other cysts*
 3. *vascular brainstem lesions esp. pontine infarctions, angiomas, or AVM*
- ✗ often associated with other motor or sensory deficits
- ✗ ↑ more common with
 1. bilateral
 2. abnormal trigeminal reflex
 3. trigeminal sensory deficits

TRIGEMINAL NEURALGIA: TREATMENT

- ✗ TN due to focal lesion compressing sensory root of V nerve – surgical decompression
- ✗ Primary TN – medical or surgical

TRIGEMINAL NEURALGIA: MEDICAL TREATMENT

- ✗ Carbamazepine (CBZ): most effective drug, ~75% respond, first-line drug
- ✗ risk of severe allergic reaction (Steven Johnson syndrome) with CBZ check HLA-B1502 before starting
- ✗ vertigo, drowsiness, ataxia are common side effects, esp elderly
- ✗ once pain controlled completely, dose tapered every few weeks to determine whether remission has developed
- ✗ regular blood counts in first few months, then yearly for agranulocytosis

TRIGEMINAL NEURALGIA: MEDICAL TREATMENT

Second-line drugs: considered for trial, alone or in combination, when CPZ is not helpful or tolerated

- ✗ Gabapentin: benign side effects, beneficial in neuropathic pain
- ✗ Pregabalin
- ✗ Phenytoin
- ✗ Baclofen
- ✗ Valproate
- ✗ Clonazepam
- ✗ Lamotrigine
- ✗ Oxcarbazepine
- ✗ Topiramate

TRIGEMINAL NEURALGIA: NONMEDICAL TREATMENT

- ✗ **Alcohol block** of peripheral branch of V nerve
- ✗ **Microvascular decompression** (separation of V nerve compressing vessel), mortality 1-2% & risk of damage to other cranial nerves [YOUNG patients]
- ✗ **Percutaneous radiofrequency thermocoagulation** of V nerve sensory root (corneal sensory loss can lead to serious eye Cx; anesthesia dolorosa, a distressingly painful sensation in the numb area, occurs occasionally [ELDERLY patients])
- ✗ Section of sensory root of V nerve (surgical) used less often now

Cluster headache and other TACs

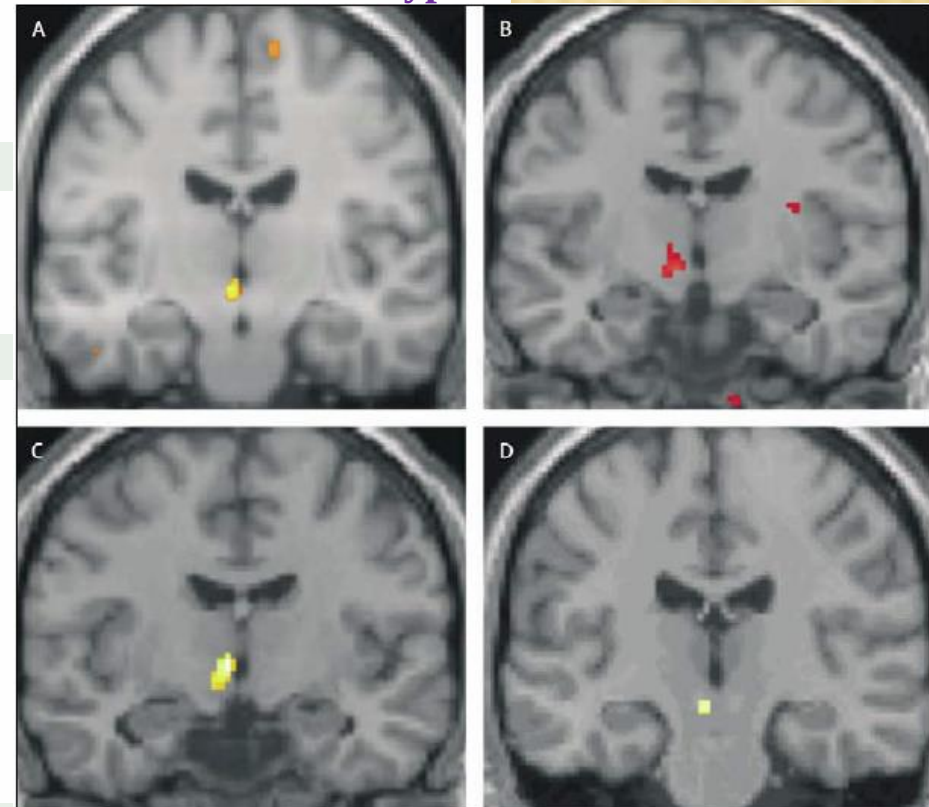
- Cluster headache
- Paroxysmal hemicrania
- Short-lasting unilateral neuralgiform headaches with conjunctival injection and tearing (SUNCT)
- Probable TAC
- Refractory TAC

Diagnostic criteria for cluster headache according to ICHD II

- A. At least five attacks fulfilling B through D
- B. Severe or very severe unilateral orbital, supraorbital and/or temporal pain lasting 15 to 180 minutes if untreated
- C. Headache is accompanied by at least one of the following:
 - 1. Ipsilateral conjunctival injection and/or lacrimation
 - 2. Ipsilateral nasal congestion and/or rhinorrhea
 - 3. Ipsilateral eyelid edema
 - 4. Ipsilateral forehead and facial sweating
 - 5. Ipsilateral miosis and/or ptosis
 - 6. A sense of restlessness or agitation
- D. Attacks have a frequency from one every other day to eight per day
- E. Not attributed to another disorder



Functional MRI during cluster headache attack reveals activation of **hypothalamus**



Epidemiology

Sex (male:female)	3:1
Prevalence	0.9%
Age of onset (years)	28–30

Pain

Quality	Piercing, throbbing
Intensity	Extremely high
Localisation	Periorbital
Duration of attack	15–120 min

Frequency of attacks	1–8 per day
Autonomic symptoms	++
Circadian rhythmicity	+
Alcohol trigger	++

Headache of Sudden Onset

Primary headache disorders

- Crash migraine
- Cluster headache
- Benign exertional headache
- Benign orgasmic cephalgia

Secondary headache disorders

Associated with vascular disorders

- Unruptured saccular aneurysm
- SAH
- Internal carotid artery dissection
- Cerebral venous thrombosis
- Acute hypertension
 - pressor response
 - phaeochromocytoma

Associated with non-vascular intracranial disorders

- Intermittent hydrocephalus
- Benign intracranial hypertension
- Pituitary apoplexy
- Infection
 - meningoencephalitis
 - acute sinusitis
- Acute mountain sickness
- Disorders of eyes
 - acute optic neuritis
 - acute glaucoma